# Rafail Psyroukis - Curriculum Vitae


**Contact Details**

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**Research Interests**

Number Theory, Modular Forms of Higher Rank (Siegel, Hermitian, Orthogonal), Algebraic Modular Forms, $L$-functions, Fourier-Jacobi Forms, Quadratic Forms, Clifford Algebras, Quaternion Algebras.

**Employment**

**University of Primorska** — Koper, Slovenia
*Assistant with PhD (Postdoctoral Researcher).* — 09/2026 - Present.

**Education**

**Durham University** — Durham, UK
*PhD in Mathematics.* — 10/2021 - 01/2026

- Supervisor: Prof. Thanasis Bouganis.
- Thesis Title: On Fourier–Jacobi Dirichlet Series for Hermitian and Orthogonal Modular Forms.

**University of Cambridge, Churchill College** — Cambridge, UK
*Master of Mathematics ("Cambridge Part III").* — 10/2020 - 07/2021

- Grade: Distinction, Rank: 29/272.
- Part III Essay (equivalent to a Master's Thesis): Two-Descent on the Jacobian of a Hyperelliptic Curve.
- Supervisor: Prof. Tom Fisher.

**University of Cambridge, Churchill College** — Cambridge, UK
*B.A. (Honours) Mathematics.* — 10/2017 - 07/2020

- Grade: First Class Honours, Rank: N/A (due to COVID-19).

**Publications and Preprints**

1. A Fourier-Jacobi Dirichlet series attached to modular forms of $\mathrm{SO}(2,4)$ (with T. Bouganis). *Submitted.* [arXiv].
2. On a Rankin-Selberg integral of three Hermitian cusp forms (with T. Bouganis). ***Ann. Math. Québec**, 50, 633-693 (2026).* [Journal], [arXiv].
3. A note on the twisted degree 6 $L$-function for Hermitian cusp forms of degree 2. ***Int. J. Number Theory**, 22, 06 (2026).* [Journal], [arXiv].
4. Analytic properties of an orthogonal Fourier-Jacobi Dirichlet series. ***Ramanujan J**, 69, 94 (2026).* [Journal], [arXiv].
5. A Fourier-Jacobi Dirichlet series for cusp forms on orthogonal groups. ***Res. number theory** 11, 90 (2025).* [Journal], [arXiv].

**Teaching/ Marking/ Mentoring**

**University of Durham.**

**Number Theory III** | Durham University — 2025 - 2026

- Marker for third year module "Number Theory III", Durham University.
- Marked weekly assignments.

**Discrete Mathematics** | Durham University 2025 - 2026

- Marker for first year module "Discrete Mathematics", Durham University.
- Marked weekly assignments.

**Algebra II** | Durham University 2022 - 2025

- Tutor and marker for the second year module "Algebra II", Durham University.
- Provided bi-weekly tutorials to groups of roughly 16 students and marked weekly assignments.

**Elementary Number Theory II** | Durham University 2023 - 2024

- Tutor and marker for the second year module "Elementary Number Theory II", Durham University.
- Provided bi-weekly tutorials to groups of roughly 16 students and marked weekly assignments.

**Linear Algebra** | Durham University 2021 - 2023

- Marker for first year module "Linear Algebra", Durham University.
- Marked weekly assignments.

**Other Activities.**

**Marker** | Cambridge University Press & Assessment 06/2026 - 07/2026

- Marked STEP 2 and STEP 3 Exam papers.

**Problem Solving Matters** | Durham University 06/2024 - 09/2024

- Mentor for students preparing for mathematics admissions tests such as STEP/MAT/TMUA.
- Program run by MEI Advanced Mathematics Support Programme.

## Talks Given

**Building Bridges, Będlewo Palace.** 18/08/2026
*"A Fourier-Jacobi Dirichlet series attached to modular forms of* $\mathrm{SO}(2,4)$*."*

**Panhellenic Conference of Mathematical Analysis, NTUA.** 21/12/2025
*"Analytic properties of a certain Dirichlet series attached to orthogonal modular forms." (**Invited**)*

**33èmes Journées Arithmétiques, University of Luxembourg.** 01/07/2025
*"A Dirichlet series attached to orthogonal modular forms".*

**PGR Research Day, Durham University.** 12/05/2025
*"From classical to Siegel modular forms".*

**Number Theory Seminar, Adam Mickiewicz University.** 04/11/2024
*"Analytic properties of Fourier-Jacobi Dirichlet series". (**Invited**)*

**Willmore Pure Postgraduate Day, Durham University.** 08/10/2024
*"Analytic Properties of Rankin-Selberg Convolutions". (**Invited**)*

**Y-RANT VI, University of Oxford.** 01/08/2024
*"A Dirichlet series for modular forms on orthogonal groups".*

**British Mathematical Colloquium, University of Manchester.** 19/06/2024
*"A Dirichlet Series attached to orthogonal modular forms".*

**PGR Research Day, Durham University.** 22/05/2024
*"Modular Forms and Applications".*

**Arithmetic Study Group, Durham University.** 30/01/2024
*"On a Rankin-Selberg integral of three Hermitian cusp forms". (**Invited**)*

**GANDALF Postgraduate Seminar, Durham University.** 06/12/2023
*"Hermitian Modular Forms and Dirichlet Series".*

**Number Theory Seminar, University of Nottingham.** 01/11/2023
*"On a Rankin-Selberg integral of three Hermitian cusp forms". (**Invited**)*

**Y-RANT V, University of Cambridge.** 07/09/2023
*"On a Rankin-Selberg integral of three Hermitian cusp forms and its relation with the twisted spinor L-function".*

## Conferences and Workshops

**Building Bridges.** 10th - 21st August 2026
*Summer School and Workshop on Automorphic Forms and Related Topics.*

**Panhellenic Conference of Mathematical Analysis.** 18th - 20th Dec. 2025
*National Technical University of Athens.*

**Heilbronn Annual Conference 2025.** 10th - 11th July 2025
*University of Bristol.*

**33èmes Journées Arithmétiques.** 30th June - 4th July 2025
*University of Luxembourg.*

**British Mathematical Colloquium.** 17th - 20th June 2024
*University of Manchester.*

**1st Greek Number Theory Meeting.** 21st - 22nd Dec. 2023
*National and Kapodistrian University of Athens.*

**Automorphic Forms and $L$-functions of higher rank.** 11th - 15th Sept. 2023
*Queen Mary University London.*

**Arithmetic of $L$-functions.** 22nd - 26th May 2023
*Instituto de Ciencias Matemáticas.*

**Northern Number Theory Seminar (NuTS).** 11th May 2023
*University of York.*

**Heilbronn Doctoral Partnership Summer Workshop.** July - August 2022
*Group work on the project "Latency in Communications", proposed by the DSTL.*

## Scholarships

**Doctoral Research Excellence Award.** 2025
*Awarded annually to a PhD student who has demonstrated excellence in research throughout their doctoral studies in Mathematical Sciences, Durham University.*

**Heilbronn Doctoral Partnership Studentship.** 2021 - 2025
*Studentship offered by the Heilbronn Institute for Mathematical Research in support of my PhD studies, grant number (EP/V521917/1).*

**Onassis Foundation Scholarship.** 2022 - 2025
*Scholarship offered to academically excellent students with Greek citizenship, in support of their research.*

**Cambridge Bursary.** 2017 - 2021

*Income-assessed bursary, maximum value offered each year.*

**Scholarship for Academic Performance.** 2019 - 2021

*Scholarship awarded to students who achieve a First (undergraduate studies) or a Distinction (master's studies) from Churchill College, University of Cambridge.*

## Professional Service

**Referee for**: Journal of Number Theory, International Journal of Number Theory.

**Reviewer for**: Mathematical Reviews.

## Awards and Honours

- **International Mathematical Olympiad (IMO)**, Bronze Medal. July 2017
- **Balkan Mathematical Olympiad (BMO)**, Bronze Medal. May 2017
- **Mediterranean Mathematical Olympiad**, Honourable Mention. April 2017
- **Greek Physics Contest**, 24th prize. May 2017
- **Greek Mathematical Olympiad**, Silver Medal (×4). Feb 2014 - 2017

## Skills

**Languages**: English, Greek (Native).

**Programming**: LaTeX, MATLAB, Python, SageMath, MAGMA.